

The q-pure-map conjecture on $M_3(\mathbb{C})$

A literature-implied full solution of the $n = 3$ subcase

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Status. Literature-implied answer (full $n = 3$ subcase), likely valid. The implication assembled below was not found stated explicitly in the cited papers. The conjecture for $n \geq 4$ remains outside the scope of this note.

1 Source conjecture and exact scope

In the concluding paragraph of arXiv:0905.2708v2, PDF page 37, Jankowski asks for all unital q-pure maps $\phi : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ and states the conjecture that every such map is invertible or has rank one [1]. The same paragraph notes that the $n = 2$ case is proved in later work and suggests using the limit map for $n = 3$.

This note proves exactly that next case.

Theorem 1 (The $n = 3$ case). *If $\phi : M_3(\mathbb{C}) \rightarrow M_3(\mathbb{C})$ is unital and q-pure, then either $\text{rank } \phi = 1$ or ϕ is invertible.*

Here q-pure has the source-paper meaning: the q-subordinates of ϕ are precisely 0 and

$$\phi^{(s)} = \phi(I + s\phi)^{-1}, \quad s \geq 0.$$

2 Identification mechanism

The proof joins two results that were developed for different purposes. The 2010 paper classifies every possible limit idempotent

$$L_\phi = \lim_{s \rightarrow \infty} s\phi(I + s\phi)^{-1}$$

for a unital q-positive map on $M_3(\mathbb{C})$, and proves that a q-positive map annihilating a nonzero positive matrix is not q-pure [2, Theorem 3.9, Corollary 3.10, Proposition 4.5]. The 2018 paper proves that the range of a finite-dimensional, index-zero, q-pure q-weight is a factor under its Choi–Effros product [3, Theorem 5.7].

The bridge is supplied by the boundary-weight double already constructed in the source paper. The next lemma records the bridge explicitly because this is the nontrivial identification on which the result depends.

Lemma 1 (Boundary-weight bridge). *Let $\phi : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ be unital and q-pure. Choose a normalized unbounded vector boundary weight ν of the form required in Lemma 4.3 of [1], and let ω be the q-weight dual to the boundary weight map of the double (ϕ, ν) . Then:*

- (i) ω is q-pure and has index zero;

(ii) $\text{ran } \omega = \text{ran } \phi$;

(iii) the Choi–Effros limit of ω is L_ϕ .

Consequently $\text{ran } \phi$, with product $A \star B = L_\phi(AB)$, is a factor.

Proof. The source paper defines the dual q -weight by

$$\omega(A) = \phi(\Omega_\nu(A)),$$

where Ω_ν applies ν entrywise. Lemma 4.3 of [1] says that every subordinate CP-flow of the flow induced by (ϕ, ν) is induced by (ψ, ν) for a q -subordinate $\psi \leq_q \phi$. Since q -purity of ϕ makes all such ψ belong to the chain $\{0\} \cup \{\phi^{(s)} : s \geq 0\}$, the q -subordinates of ω are totally ordered. Thus ω is q -pure in the sense of [3, Definition 1.2]. Corollary 3.3 of [1] gives index zero.

Normalization gives $\nu(I - \Lambda(1)) = 1$. For every $B = (b_{ij}) \in M_n(\mathbb{C})$, the matrix $A = (b_{ij}(I - \Lambda(1)))$ belongs to the null boundary algebra and satisfies $\Omega_\nu(A) = B$. Hence Ω_ν is onto and $\text{ran } \omega = \text{ran } \phi$.

For $t > 0$, put $s_t = \nu_t(\Lambda(1))$. Equation (3) in [1] gives the generalized boundary representation

$$\pi_t^\# = \phi(I + s_t \phi)^{-1} \Omega_{\nu_t}.$$

Since $\Omega_{\nu_t}(\Lambda(B)) = s_t B$,

$$\pi_t^\# \Lambda(B) = s_t \phi(I + s_t \phi)^{-1}(B).$$

Unboundedness of ν implies $s_t \rightarrow \infty$ as $t \downarrow 0$, so the right-hand side converges to $L_\phi(B)$. This proves (iii). The final assertion is now exactly Theorem 5.7 of [3]. $\square$

3 Proof of Theorem 1

Proof. For a unital q -positive ϕ on $M_3(\mathbb{C})$, the 2010 limit-map lemma shows that L_ϕ is a unital completely positive idempotent and that

$$\text{ran } L_\phi = \text{ran } \phi, \quad \ker L_\phi = \ker \phi.$$

Theorem 3.9 of [2] classifies such idempotents up to unitary conjugacy. Proposition 4.5 of the same paper excludes every case in which L_ϕ annihilates a nonzero positive matrix: equality of kernels would make ϕ annihilate that matrix, contradicting q -purity.

Thus, apart from a faithful rank-one map, the only possible classified forms for L_ϕ are the following four forms (the labels are those of [2]):

$$\begin{aligned} \text{(IV)} \quad & L(A) = \text{diag}(a_{11}, a_{22}, a_{33}), & \text{ran } L &\cong \mathbb{C}^3, \\ \text{(V)} \quad & L(A) = a_{11} \oplus \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}, & \text{ran } L &\cong \mathbb{C} \oplus M_2(\mathbb{C}), \\ \text{(VI)} \quad & L(A) = a_{11} \oplus bI_2, \quad b = \lambda a_{22} + (1 - \lambda)a_{33}, & \text{ran } L &\cong \mathbb{C} \oplus \mathbb{C}, \\ \text{(VII)} \quad & L(A) = A, & \text{ran } L &= M_3(\mathbb{C}). \end{aligned}$$

In cases (IV)–(VI), the range is already closed under ordinary matrix multiplication and L fixes every range element. Therefore its Choi–Effros product $X \star Y = L(XY)$ is ordinary multiplication. Each of $\mathbb{C}^3$, $\mathbb{C} \oplus M_2(\mathbb{C})$, and $\mathbb{C} \oplus \mathbb{C}$ has nontrivial center, so none is a factor. Lemma 1 eliminates these three cases.

If L_ϕ is the faithful rank-one form, equality of ranges gives $\text{rank } \phi = 1$. The only remaining case is (VII), where $\text{ran } \phi = \text{ran } L_\phi = M_3(\mathbb{C})$. Hence ϕ is a surjective linear endomorphism of the nine-dimensional vector space $M_3(\mathbb{C})$ and is therefore invertible. This proves the theorem. $\square$

4 Verification, limitations, and novelty check

- **Internal consistency.** The range calculation uses an explicit preimage under Ω_ν , and the limit calculation follows directly from the source formula for $\pi_t^\#$. No computational or asymptotic approximation is used.
- **Classification check.** The three excluded nonfactor ranges are respectively $\mathbb{C}^3$, $\mathbb{C} \oplus M_2(\mathbb{C})$, and $\mathbb{C}^2$; the two surviving factor ranges are $\mathbb{C}$ and $M_3(\mathbb{C})$.
- **Scope.** The proof settles the full $n = 3$ subcase only. It provides no classification for $n \geq 4$.
- **Bounded search.** The four lightweight run indexes, the exact conjecture wording, the phrases “q-pure M_3 ” and “unital q-pure maps,” and the arXiv records/citation trail for 0905.2708, 1005.4404, 1106.2304, and 1807.09824 were checked. The explicit $n = 2$ result and the later q-weight classification were found; no paper explicitly stating the $n = 3$ matrix-map corollary was located. Novelty confidence is therefore moderate, and the result is labeled literature-implied rather than wholly new.
- **Human-review focus.** Verify that Lemma 4.3 of the source paper covers every q-weight subordinate under the standard CP-flow/q-weight correspondence. Once that order transfer is accepted, the remaining steps are formal consequences of the cited theorems.

References

- [1] C. Jankowski, *On type II_0 E_0 -semigroups induced by boundary weight doubles*, arXiv:0905.2708v2. Source conjecture on PDF p. 37; boundary-weight construction, Corollary 3.3 and Lemma 4.3.
- [2] C. Jankowski, *Unital q-positive maps on $M_2(\mathbb{C})$ and a related E_0 -semigroup result*, arXiv:1005.4404v1. See Theorem 3.9 and Corollary 3.10 on PDF p. 20, and Proposition 4.5 on PDF p. 23.
- [3] C. Jankowski, D. Markiewicz, and R. T. Powers, *Classification of q-pure q-weight maps over finite dimensional Hilbert spaces*, arXiv:1807.09824v1. See Definition 1.2 and Theorem 5.7 on PDF p. 44.